

ECON726: Probability Homework

Instructions: Please submit a **handwritten** copy of your answers. When using any calculations, please show all of your work.

1. Suppose you have an unfair, two-sided coin. The probability of flipping heads is $\frac{1}{3}$. What is the probability of flipping tails?
2. In a standard deck of 52 cards, what is the probability of drawing a red card or a face card (Jack, Queen, King)?
3. Suppose a random variable X follows a normal distribution with mean μ and a standard deviation σ . What is the probability that X falls within two standard deviations of the mean?
4. Suppose you are selecting a card from a standard deck of 52 cards. What is the probability that the card is a spade, given that the card is black?
5. I have a new AI technology that predicts whether street signs are damaged. Let:
 - P denote whether the street sign is **predicted** as damaged:
 - $P = 1$: the technology predicts the sign is damaged
 - $P = 0$: the technology predicts the sign is not damaged
 - D be the **ground truth**; whether the sign is actually damaged:
 - $D = 0$: the sign is not damaged
 - $D = 1$: the sign is damaged

Further, suppose:

- The **prevalence** of damaged signs is $\frac{1}{100}$:

$$\mathbb{P}(D = 1) = \frac{1}{100}$$

- The **False Positive Rate** is the probability that the AI predicts a sign is damaged when it isn't:

$$\mathbb{P}(P = 1|D = 0) = \frac{1}{50} = 0.02$$

- The **False Negative Rate** is the probability that a sign is predicted as not damaged, when it actually is damaged. Suppose this equals zero:

$$\mathbb{P}(P = 0|D = 1) = 0$$

- (a) Find $\mathbb{P}(P = 1|D = 1)$. This is the probability that actually damaged signs are predicted damaged, known as the **True Positive Rate**.

- (b) Find $\mathbb{P}(P = 1)$ (Hint: use the Law of Total Expectation)
- (c) Find the probability that a sign predicted damaged is actually damaged: $\mathbb{P}(D = 1|P = 1)$
(Hint: use Bayes' Theorem)
- (d) Based on your calculations above, do you think the NYC should invest in this technology?